

STAT 331 - PRACTICE ASSIGNMENT 2
WINTER 2026
Due: 3:00pm Feb 27 (Friday), 2026

Reminder:

- This practice assignment is designed to support your learning. It is **completion-based**, and you will receive **1 participation credit** if you:
 - submit the assignment on time;
 - attempt all questions and include all R code and relevant output used in your work; and
 - submit original work.
- Submit your work by the due time via Crowdmark. Please ensure that your work is legible and that all pages are oriented correctly.

1. Consider the simple linear regression model:

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2), \quad i = 1, 2, \dots, n$$

- (a) Re-write this model in vector/matrix form. Explicitly define the response vector $\vec{Y}$, the design matrix X , the regression coefficient vector $\vec{\beta}$, and the random error vector $\vec{\epsilon}$, including their dimensions and entries.
- (b) Using matrix algebra and the least squares formula $\tilde{\vec{\beta}} = (X^T X)^{-1} X^T \vec{Y}$, derive explicit closed-form expressions for the least squares estimators $\tilde{\beta}_0$ and $\tilde{\beta}_1$, as well as their variances and covariance. Your final expressions should reduce to, and agree with, the standard formulas previously derived for the simple linear regression model.
- (c) Show that the least squares estimator

$$\tilde{\vec{\beta}} = (X^T X)^{-1} X^T \vec{Y}$$

is the best linear unbiased estimator (BLUE) for $\vec{\beta}$. That is, consider any other linear unbiased estimator of the form

$$\tilde{\vec{\beta}}^* = [A + (X^T X)^{-1} X^T] \vec{Y}$$

where A is a constant matrix.

- (i) Determine the condition(s) on A required for $\tilde{\vec{\beta}}^*$ to be unbiased.
- (ii) Under the condition(s) in (i), show that

$$\text{Var}(\tilde{\beta}_j^*) \geq \text{Var}(\tilde{\beta}_j) \quad \text{for each component } \beta_j \text{ of } \vec{\beta}.$$

2. Suppose the following multiple linear regression model is fit to 15 observations:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \quad i = 1, 2, \dots, 15.$$

The following summaries are available:

$$(X^T X)^{-1} = \begin{bmatrix} 2 & -5 & -3 \\ -5 & 1 & -1 \\ -3 & -1 & 5 \end{bmatrix} \quad \text{and} \quad \hat{\vec{\beta}} = \begin{bmatrix} 52 \\ 10 \\ 3 \end{bmatrix} \quad \text{and} \quad SSE = \sum_{i=1}^{15} r_i^2 = 30 .$$

Using these results, answer the following questions:

- (a) Fitted model: Write the equation for the fitted value $\hat{\mu}_i$ in terms of the estimated coefficients.
 - (b) Estimate of random error variance: compute $\hat{\sigma}^2$.
 - (c) Compute the estimated variance $\widehat{Var}(\hat{\beta}_1)$.
 - (d) Hypothesis test for β_1 : Test $H_0 : \beta_1 = 0$ v.s. $H_a : \beta_1 \neq 0$ at the 5% significance level. State the value of the test statistic and clearly report your conclusion.
3. Consider a multiple linear regression model with an intercept and p explanatory variables. Let $\vec{y}$ be the $n \times 1$ vector of observed responses, X the $n \times (p+1)$ design matrix (including a column of ones for the intercept), and $\hat{\vec{\beta}} = (X^T X)^{-1} X^T \vec{y}$ the $(p+1) \times 1$ vector of least squares estimates.

Let $\hat{\vec{\mu}} = X \hat{\vec{\beta}}$ be the vector of fitted values, and $\vec{r} = \vec{y} - \hat{\vec{\mu}}$ the residual vector.

Show that the following properties hold in general:

- (a) Sum of responses equals sum of fitted values:

$$\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{\mu}_i$$

- (b) Residuals are orthogonal to fitted values:

$$\hat{\vec{\mu}}^T \vec{r} = 0$$

- (c) R-square as squared correlation:

The coefficient of determination, R^2 , is equal to the square of the sample correlation between observed responses y_i and fitted values $\hat{\mu}_i$:

$$\rho = \frac{\sum (y_i - \bar{y})(\hat{\mu}_i - \bar{\hat{\mu}})}{\sqrt{(\sum (y_i - \bar{y})^2) (\sum (\hat{\mu}_i - \bar{\hat{\mu}})^2)}} .$$

(This is why R^2 is also called multiple correlation coefficient.)

4. The quality (Y) of Pinot Noir wine is thought to be related to its clarity (x_1), aroma (x_2), body (x_3), flavour (x_4), and oakiness (x_5). Data for 38 wines are given in the data file “`wine.txt`”.

Answer the following questions:

- Full model: Fit a multiple linear regression model relating wine quality to all five explanatory variables. Obtain the least squares estimates for the regression coefficients and write the fitted equation. Interpret the estimated effect of each explanatory variable.
 - Significance of coefficients: Use t -tests to evaluate the significance of each regression coefficient at the $\alpha = 0.05$ level. Discuss your finding, including which explanatory variable appear most important in determining wine quality.
 - Reduced model and prediction: Fit a reduced model including only **Aroma**, **Flavour** and **Oakiness**. Using this model, predict the quality of a new Pinot Noir wine with the properties: **Aroma**= 6, **Flavour**=7 and **Oakiness**=3. Provide the predicted quality and a 95% prediction interval.
 - Model evaluation: for the reduced model in (c), perform an ANOVA F -test to evaluate whether the model explains a significant portion of variation in wine quality. State the F -statistics, degrees of freedom, and p -value, and draw your conclusion.
 - Compute the coefficient of determination (R^2) for this model and interpret its meaning.
5. A consumer-product marketing experiment involved different advertising and promotion strategies in eight sales regions. TV advertising, newspaper advertising, and retailer incentive (absent or present) were systematically varied. TV advertising was either reduced by 10% compared with the previous quarter, kept the same, or increased by 10%, thus coded as -1 , 0 , or 1 . Similar changes were made to the newspaper advertising, again coded as -1 , 0 , or 1 . The response is the change in market share (percentage points) relative to the previous quarter. Data are given in the table below.

TV advertising	Newspaper advertising	Retailer incentive	Change in market share
-1	0	Absent	1.2
0	-1	Absent	1.5
0	1	Absent	2.2
1	0	Absent	2.3
-1	0	Present	1.9
0	-1	Present	1.4
0	1	Present	2.8
1	0	Present	2.3

Consider a multiple linear model for market share change, including TV advertising, newspaper advertising and retailer incentive as explanatory variables.

- Specify the model: Write the multiple linear regression model, make sure to introduce all the necessary notation. Fit the model in R use the provided data, and provide the fit summary.

- (b) Do any of the strategies (TV advertising, newspaper advertising, and retailer incentive) have a significant effect on the change in market share, while controlling for the other strategies?
- Perform the appropriate hypothesis tests for each regression coefficient.
 - Provide justification for your conclusions based on the test results (e.g., t -statistics, p -values).
 - For any strategy that is significant, interpret the estimated effect in the context of the problem (i.e., how a one-unit change in the strategy affects market share change, holding other strategies constant).
- (c) Company-wide, one unit of TV advertising (on the coded scale) would cost \$3,250,000 and one unit of newspaper advertising would cost \$2,600,000. The Vice President for Marketing wants to know which advertising strategy, TV or newspaper, is more cost effective. To answer this question, estimate mean market share change per million dollars spent in TV advertising (while holding other factors fixed), and provide a 95% CI. Do the same for newspaper advertising. Compare these results and explain your findings in words.
- (d) Based on your analysis, the Vice President for Marketing is considering the following company-wide strategy: keep TV advertising at level 0 (on the coded scale), increase newspaper advertising by 2 unit, and do not use the retailer incentive. Predict the market-share change for a new region that will adopt this strategy, and provide a 95% prediction interval.
6. Many factors may affect how much health insurance premiums cost. The data file “insurance.txt” on the LEARN course website contains information from a large sample of insurance contracts, including following variables
- **age**: age of primary beneficiary
 - **sex**: insurance contractor gender (female, male)
 - **bmi**: Body mass index
 - **children**: Number of children covered by health insurance
 - **smoker**: Smoking (yes, no)
 - **region**: The beneficiary’s residential area in the US (northeast, northwest, southeast, southwest,)
 - **charges**: Individual medical costs billed by health insurance

Acknowledgements: The dataset is available on GitHub.

- (a) Consider a multiple linear model, medical insurance cost being the response and all the other variables being the predictors. Provide the form of the model, carefully define the indicators for categorical variables. Fit the model in R and provide a fit summary.
- (b) Now focus on regional differences. Is there evidence of differences in insurance charges between the following pairs of regions, after controlling for other variables in the model?

- i) Northeast vs. Northwest
- ii) Southeast vs. Southwest
- iii) Northeast vs. Southeast
- iv) Northwest vs. Southwest

For each comparison:

- Clearly state the null and alternative hypotheses in terms of the relevant regression coefficient(s).
 - Specify the test statistic you use.
 - State your conclusion in context.
- (c) Is region an important predictor of health insurance costs? Conduct a test for the overall effect of region in the regression model. In your answer:
- Clearly state the null and alternative hypotheses in terms of the relevant regression coefficient(s).
 - Specify the test statistic you use.
 - State your conclusion in context.